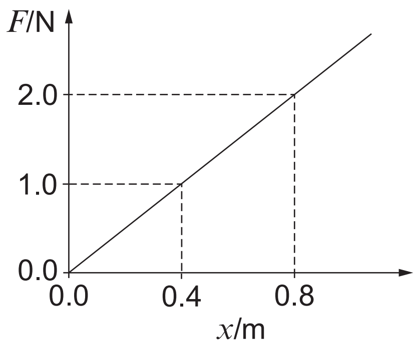
Examiner only

2. (a) The graph shows how the force, F, needed to stretch a simple spring varies with the extension, x, of the spring. The spring obeys Hooke's law.



(i) Calculate the work done in stretching the spring 0.4 m from its unstretched position. [2]

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(ii) Explain why the work done to stretch the spring 0.8 m from its unstretched position is **not twice** your answer to part (a)(i). [2]

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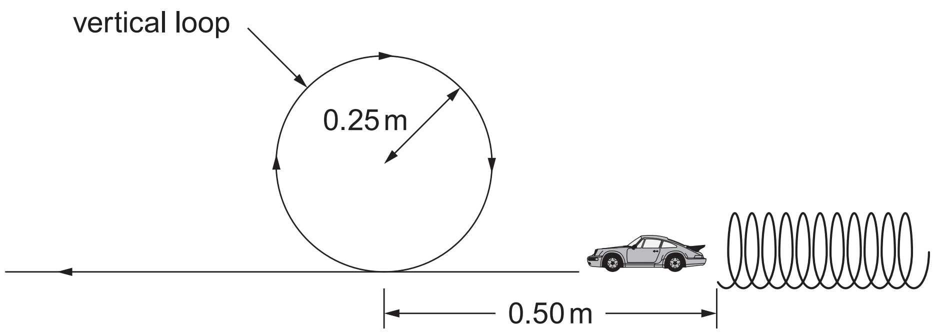
(b) The spring in part (a) is used to propel a toy car of mass 30 g along a track which contains a vertical loop of radius 0.25 m.



Examiner only

(i) The spring is compressed by 0.4 m. The car is placed against the end of the spring, which is then released. Assuming that all of the spring's energy is transferred to the car and that no resistive forces act on it, calculate the speed of the car at the top of the loop. [3]

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(ii) In reality resistive forces act on the car to the extent that the car loses 5% of its initial energy to frictional forces in moving from the spring to the top of the loop. Calculate the mean resistive force acting on the car during this motion. [3]

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